Experimental Determination of the Velocity of Sound at S.T.P Using Arduino

Name: Ritiz Shrestha

Organization/Club: SERN

Post: Founding Member

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1. Abstract

This project presents an Arduino-based experimental system designed to determine the speed of sound in air using ultrasonic sensing. A HC-SR04 ultrasonic sensor was used to measure the round-trip travel time of an ultrasonic pulse reflected from a fixed surface. A DHT11 temperature sensor was incorporated to measure the temperature, while a 16×2 I²C LCD was used to display the measured temperature and calculated speed.

The experimental speed of sound was calculated using the measured distance and ultrasonic travel time. Multiple readings were taken under approximately constant environmental conditions, and their average was determined. Since the speed of sound depends on temperature, the experimentally obtained value was converted to its equivalent value at STP and compared with the accepted value of 331.1 m/s. The percentage error was then calculated to evaluate the accuracy of the experimental setup.

This project demonstrates the practical integration of physics, electronics, sensors, microcontroller programming, and experimental data analysis into a single system.

2. Objectives

The main objectives of this project were:

1. To experimentally determine the speed of sound in air using an ultrasonic sensor.
2. To understand the time-of-flight principle used by the HC-SR04 ultrasonic sensor.
3. To measure ambient temperature using a DHT11 sensor. .
4. To calculate the speed of sound from experimental measurements.
5. To convert the experimentally measured speed to its equivalent value at STP.

6. To compare the experimental result with the accepted value of 331.1 m/s.
7. To analyze the sources of error and limitations of a low-cost experimental setup.

3. Introduction

The speed of sound is an important physical quantity in the study of waves and acoustics. In air, its value depends mainly on temperature, increasing as the temperature increases.

In this project, instead of using a conventional manual timing method, an HC-SR04 ultrasonic sensor was used to measure the travel time of an ultrasonic pulse. The sensor sends an ultrasonic wave toward a reflecting surface and detects the returning echo. The Arduino measures the time interval and uses it to calculate the speed of sound.

The project was designed to combine a theoretical physics concept with practical electronics and programming. The addition of the DHT11 temperature sensor allowed the effect of temperature on he speed of sound to be considered during the analysis.

4. Components Used

Component	Quantity	Purpose
Arduino UNO	1	Main controller and data processing
HC-SR04 Ultrasonic Sensor	1	Measurement of ultrasonic travel time
DHT11 Temperature Sensor	1	Measurement of ambient temperature
16×2 I²C LCD	1	Display of measurements
Breadboard	1	Circuit assembly
Jumper Wires	As required	Electrical connections
Reflecting Surface	1	Reflection of ultrasonic waves

5. Theory and Working Principle

The HC-SR04 works using the time-of-flight principle. When a trigger signal is applied, the sensor emits an ultrasonic pulse. The pulse travels toward a reflecting surface and returns to the sensor as an echo.

If the distance between the sensor and the reflecting surface is d, the ultrasonic wave travels a total distance of: 2d

If the measured round-trip time is t, the experimental speed of sound is:

$$v = \frac{2d}{t}$$

where:

- v = experimental speed of sound
- d = distance between sensor and reflecting surface
- t = round-trip travel time

The factor of 2 is required because the measured time corresponds to the wave travelling to the obstacle and back.

Since the speed of sound depends on temperature, the experimentally measured speed at temperature T was converted to its approximate value at 0∘C using:

$$v_0 = v_T\sqrt{\left(\frac{273}{T+273}\right)}$$

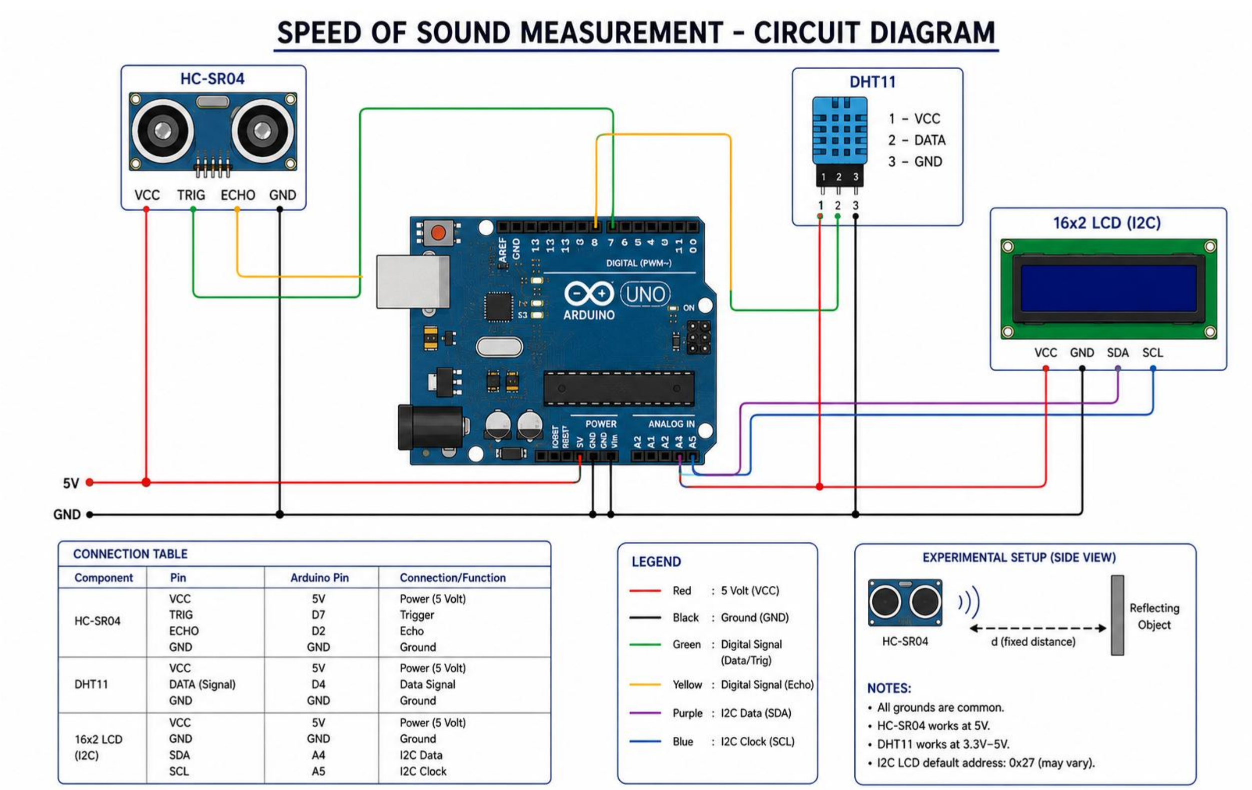
where:

- v_T = experimentally measured speed at temperature T
- v_0 = calculated speed at 0°C
- T = measured temperature in °C

The accepted speed of sound at STP was taken as: 331.1 m/s

6. Circuit Design

The system consists of three main sensing/display components connected to an Arduino UNO as shown in the figure below.



7. Experimental Setup and Procedure

The HC-SR04 was placed at a fixed distance from a flat reflecting surface. The sensor was positioned toward the surface so that the ultrasonic pulse could be reflected back toward the receiver. The DHT11 was used to measure the surrounding temperature. The Arduino processed the ultrasonic travel time and temperature data and displayed the corresponding values on the LCD.

The experiment was performed as follows:

1. The circuit was assembled according to the circuit diagram.
2. The Arduino program was uploaded and the sensors were initialized.
3. The distance between the HC-SR04 and reflecting surface was kept approximately constant.
4. The HC-SR04 measured the round-trip travel time of the ultrasonic pulse.
5. The DHT11 measured the ambient temperature.
6. The Arduino calculated the experimental speed using $v=2d/t$.
7. The temperature and calculated speed were displayed on the LCD.
8. Multiple readings were recorded under approximately the same environmental conditions.
9. The average temperature and experimental speed were calculated.
10. The experimental value was converted to STP and compared with 331.1 m/s.

8. Experimental Data

Distance between sensor and reflecting surface: approximately 29 cm

Reading	Temperature (°C)	Experimental Speed (m/s)
1	27.0	334.47
2	27.0	345.11
3	27.0	335.42
4	27.0	343.62
5	27.0	344.91
6	26.9	339.27
7	27.0	343.95
Average	**26.985°C**	**340.96 m/s**

9. Data Analysis and Calculations

Average Temperature

T_{avg}= 26.985°C

Average Experimental Speed

v_T=340.96 m/s

Conversion to STP

Using:

$v_0 = v_T\sqrt{\left(\frac{273}{T+273}\right)} \rightarrow v_0 = 323.65$ m/s

The accepted value at STP is:

$v_{accepted}$=331.1 m/s

Percentage Error

% Error = $\frac{(|331.1-323.65|)}{331.1}$ x 100%

→ % Error= 2.24%

10. Results and Discussion

The experimental system successfully measured the speed of an ultrasonic wave in air using the HC-SR04 sensor. Multiple readings were taken to reduce the influence of random variations in individual measurements.

The experimentally measured speed was obtained at approximately 29.986°C and was subsequently converted to its equivalent value at STP. This value was compared with the accepted speed of sound of 331.1 m/s.

The difference between the experimental and accepted values can be attributed to several factors, including the limited precision of the HC-SR04, uncertainty in the measured distance, the very short ultrasonic travel time, sensor alignment, and the accuracy of the DHT11 temperature measurement.

11. Sources of Error and Limitations

The major sources of uncertainty in the experiment include:

- Limited timing and distance resolution of the HC-SR04.
- Small uncertainty in measuring the sensor-to-obstacle distance.
- Very short round-trip travel time.
- Imperfect alignment of the sensor and reflecting surface.
- Temperature measurement uncertainty of the DHT11.
- Reflections from nearby objects or surfaces.
- Environmental conditions such as air movement and humidity.

12. Conclusion

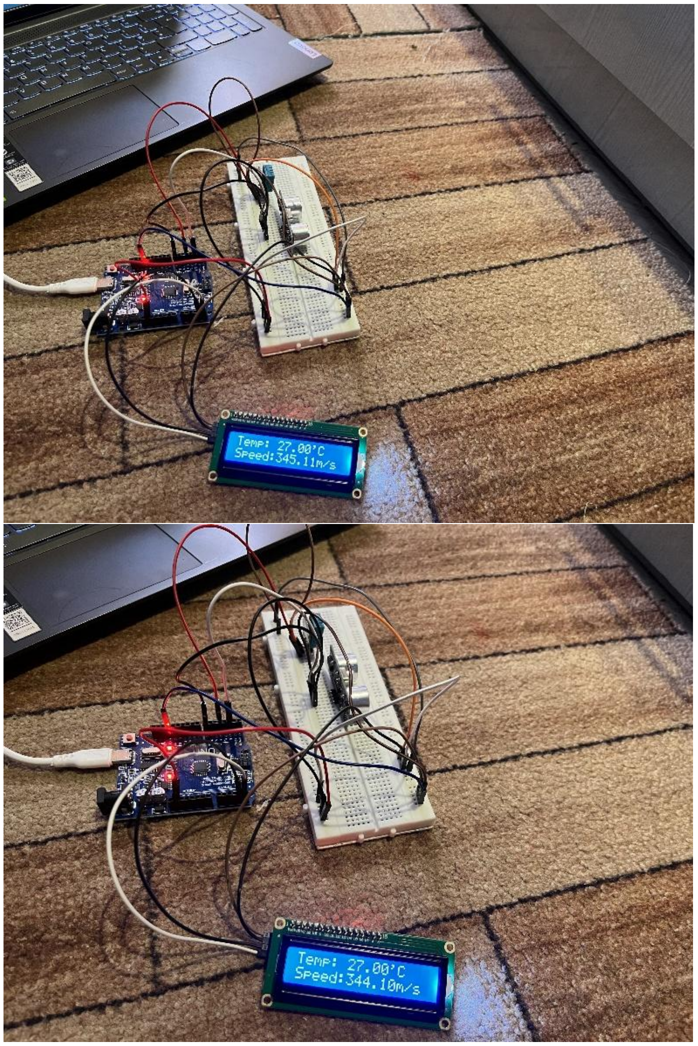
An Arduino-based experimental system was successfully developed to measure the speed of sound using ultrasonic time-of-flight measurement. The system combined a HC-SR04 ultrasonic sensor, DHT11 temperature sensor, Arduino UNO, and I²C LCD into a single functional setup.

The experimental speed was calculated from the measured ultrasonic travel time and fixed distance. The result was then converted to its equivalent value at STP and compared with the accepted value of 331.1 m/s**.**

Despite the limitations of the low-cost sensors and short measurement distance, the experiment provided a practical demonstration of the relationship between wave physics, temperature, ultrasonic sensing, electronics, and programming**.**

13. Project Evidence

Figure 3. LCD displaying experimental measurements.



Source Code (C++):

```cpp
#include<LiquidCrystal_I2C.h>
#include<DHT.h>
DHT dht(4,DHT11);
LiquidCrystal_I2C lcd(0x27,16,2);
void setup() {
  lcd.init();
  lcd.backlight();
  dht.begin();
  pinMode(7,OUTPUT);
  pinMode(2,INPUT);
}

void loop() {
  float t= dht.readTemperature();
  digitalWrite(7,LOW);
  delayMicroseconds(2);
  digitalWrite(7,HIGH);
  delayMicroseconds(10);
  digitalWrite(7,LOW);
  long time= pulseIn(2,HIGH,30000);
  float velocity = (2*29.3)/time;
  float v= velocity*10000.0;
  lcd.setCursor(0,0);
  lcd.print("Temp: ");
  lcd.print(t);
  lcd.print("'C");
  lcd.setCursor(0,1);
  lcd.print("Speed:");
  lcd.print(v);
  lcd.print("m/s");
  delay(5000);
}
```


A project by:

Ritiz Shrestha